

LAB EXPERIMENT RECORD

SET 14: Survey Results

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	10 october 2026
Day	thursday

Aim

To visualize the geographic distribution of cities using population and temperature information.

Dataset

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

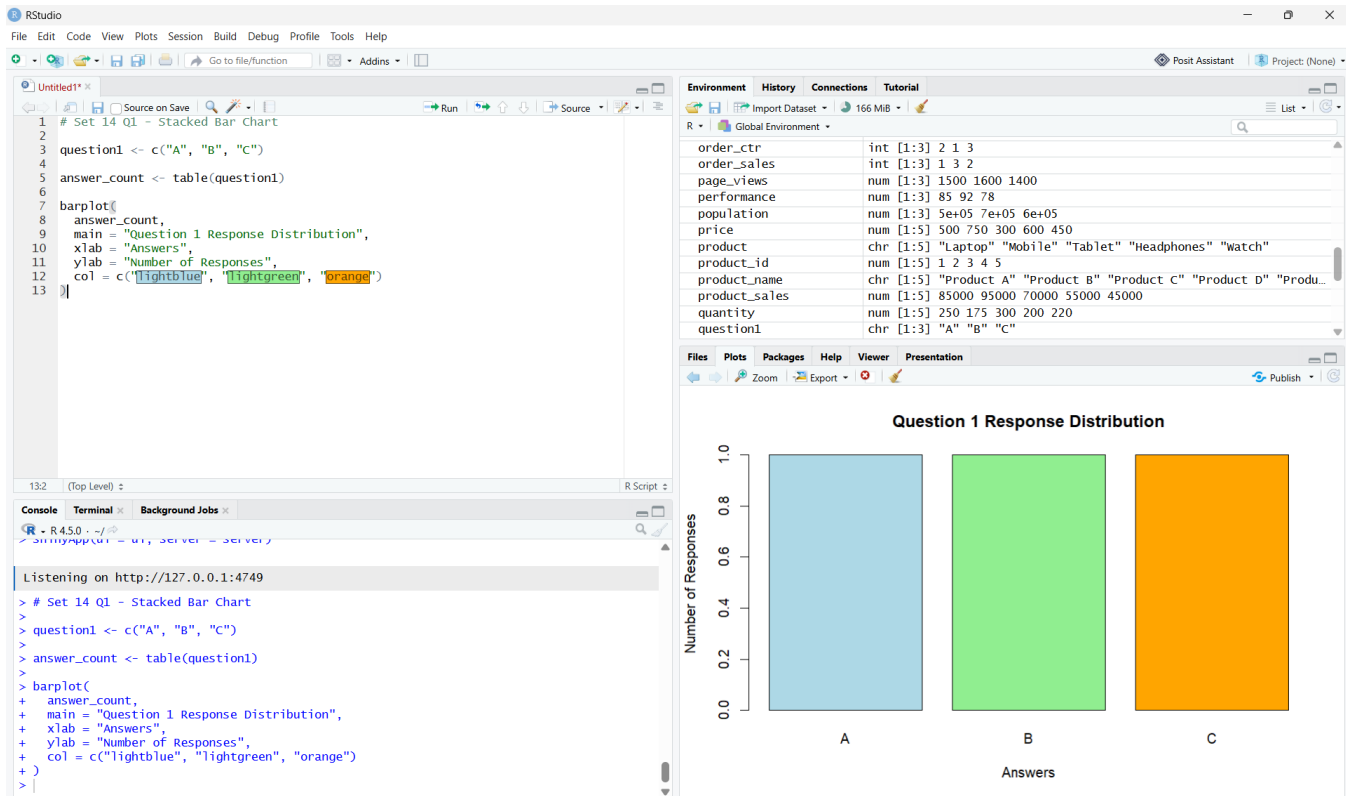
PROGRAM 1: Stacked Bar Chart

To visualize the distribution of answers for Question 1 using a stacked bar chart.

```
# # # Set 14 Q1 - Stacked Bar Chart
question1 <- c("A", "B", "C")

answer_count <- table(question1)

barplot(
  answer_count,
  main = "Question 1 Response Distribution",
  xlab = "Answers",
  ylab = "Number of Responses",
  col = c("lightblue", "lightgreen", "orange")
)
```



Observation:

- A = 1 response
- B = 1 response
- C = 1 response

Each answer has an equal frequency.

PROGRAM 2: Radar Chart

Base R does not have a built-in radar chart, so we can create a **simple radar-style chart** using Base R.

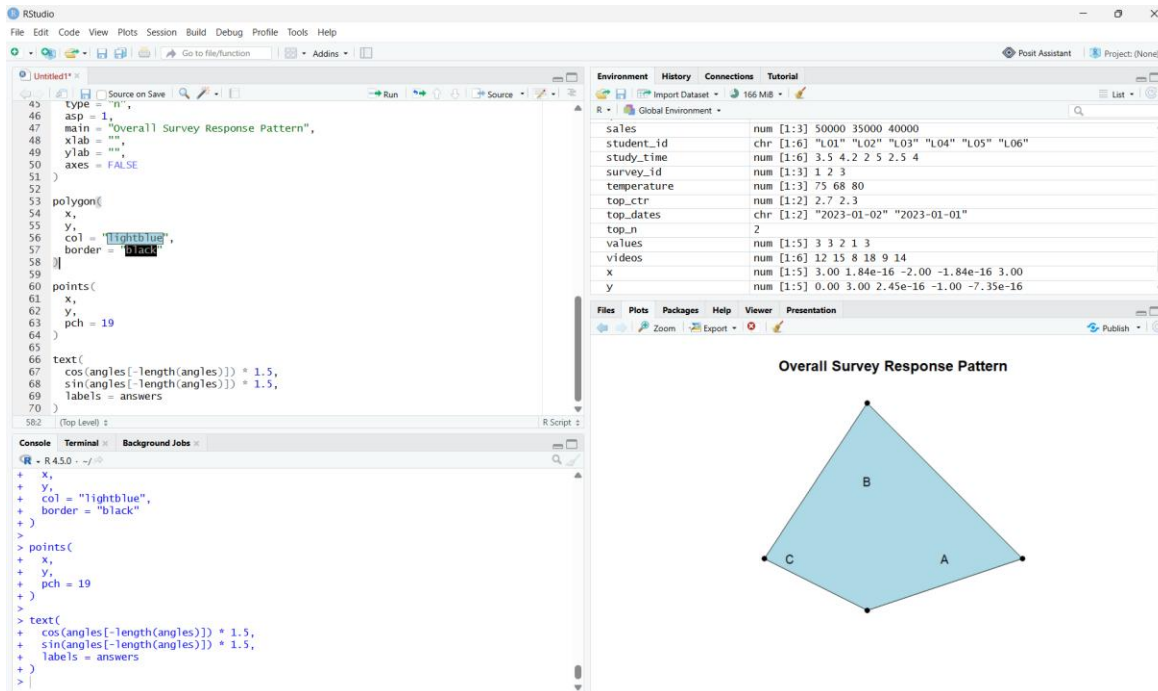
First calculate the total number of A, B, C and D responses for all questions.

```
# Set 14 Q2 - Radar Chart
question1 <- c("A", "B", "C")
question2 <- c("B", "A", "A")
question3 <- c("C", "D", "B")
answers <- c("A", "B", "C", "D")
q1 <- table(factor(question1, levels = answers))
q2 <- table(factor(question2, levels = answers))
```

```

q3 <- table(factor(question3, levels = answers))
overall <- q1 + q2 + q3
values <- as.numeric(overall)
angles <- seq(
  0,
  2 * pi,
  length.out = length(values) + 1
)
values <- c(values, values[1])
x <- values * cos(angles)
y <- values * sin(angles)
plot(
  x,
  y,
  type = "n",
  asp = 1,
  main = "Overall Survey Response Pattern",
  xlab = "",
  ylab = "",
  axes = FALSE
)
polygon(
  x,
  y,
  col = "lightblue",
  border = "black"
)
points(
  x,
  y,
  pch = 19
)
text(
  cos(angles[-length(angles)]) * 1.5,
  sin(angles[-length(angles)]) * 1.5,
  labels = answers

```



Observation:

The radar-style chart shows the overall response pattern across A, B, C and D..

PROGRAM 3: Survey Response Table

To display the survey response data for each respondent.

```
# # Set 14 Q3 - Survey Response Table

respondent <- c(1, 2, 3)

question1 <- c("A", "B", "C")
question2 <- c("B", "A", "A")
question3 <- c("C", "D", "B")

survey_table <- data.frame(
  Respondent = respondent,
  Question_1 = question1,
  Question_2 = question2,
  Question_3 = question3
)

survey_table
```

```

62 y,
63 pch = 19
64 )
65
66 text(
67   cos(angles[-length(angles)]) * 1.5,
68   sin(angles[-length(angles)]) * 1.5,
69   labels = answers
70 )
71
72
73 # Set 14 Q3 - Survey Response Table
74
75 respondent <- c(1, 2, 3)
76 question1 <- c("A", "B", "C")
77 question2 <- c("B", "A", "A")
78 question3 <- c("C", "D", "B")
79
80 survey_table <- data.frame(
81   Respondent = respondent,
82   Question_1 = question1,
83   Question_2 = question2,
84   Question_3 = question3
85 )
86
87 survey_table

```

69:19 (Top Level) ↕

Console **Terminal** **Background Jobs**

```

R - R 4.5.0 - ~/
> respondent <- c(1, 2, 3)
> question1 <- c("A", "B", "C")
> question2 <- c("B", "A", "A")
> question3 <- c("C", "D", "B")
>
> survey_table <- data.frame(
+   Respondent = respondent,
+   Question_1 = question1,
+   Question_2 = question2,
+   Question_3 = question3
+ )
>
> survey_table
  Respondent Question_1 Question_2 Question_3
1          1          A          B          C
2          2          B          A          D
3          3          C          A          B
> |

```

Observation

The survey responses of all respondents were successfully displayed in tabular form

PROGRAM 4: Survey Dashboard

```
library(shiny)

question1 <- c("A", "B", "C")
question2 <- c("B", "A", "A")
question3 <- c("C", "D", "B")

plot(
  x,
  y,
  type = "n",
  asp = 1,
  main = "Overall Survey Response Pattern",
  axes = FALSE
)

polygon(
  x,
  y,
  col = "lightblue"
)

points(
  x,
  y,
  pch = 19
)

text(
  cos(angles[-length(angles)]) * 1.5,
  sin(angles[-length(angles)]) * 1.5,
  labels = answers
)
})
```

```

output$table <- renderTable({

  data.frame(

    Respondent = 1:3,

    Question_1 = question1,

    Question_2 = question2,

    Question_3 = question3

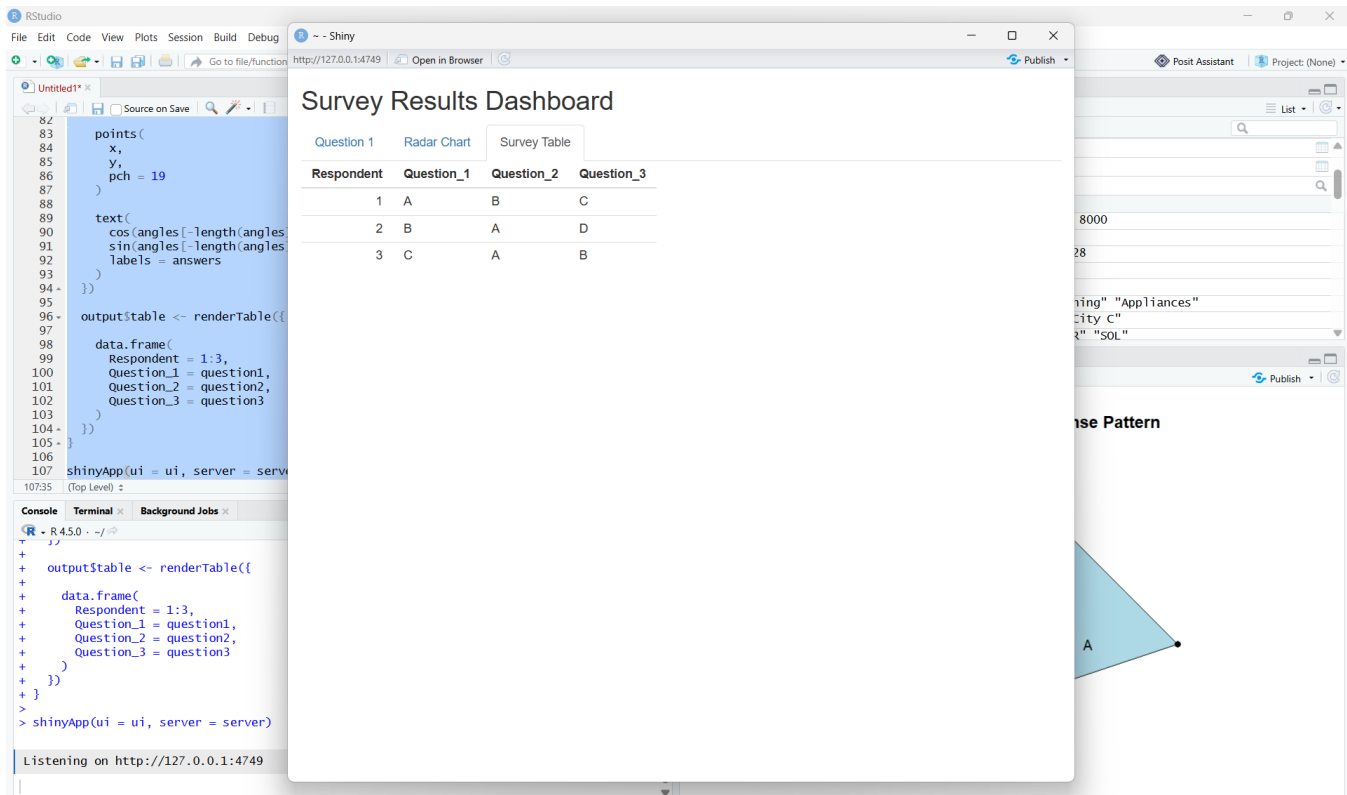
  )

})

}

shinyApp(ui = ui, server = server)shinyApp(ui = ui, server = server))

```



Output 4: Interactive Monthly Sales Dashboard

Observation: An interactive geographic dashboard was successfully developed in RStudio using Shiny.

Result:

The product category sales data was successfully analyzed using a funnel-style chart, pie chart, and table. An An interactive survey dashboard combining the stacked bar chart, radar-style chart and survey table was successfully developed using RStudio Shiny.